

Applied Statistics and Econometrics with Stata

A Practical Book of Stata Do-Files, Results, and Interpretation

Step-by-step Stata commands for statistical and econometric analysis, with
do-files, output, interpretation, and applied examples.

Prepared by
Chowdhury Amir Abdullah

Compiled in L^AT_EX

Preface

This book is a practical learning document for statistics and econometrics using Stata.

The main purpose of this book is to show how different statistical and econometric methods can be applied in Stata. Each chapter connects the method, the Stata do-file, the Stata output, and the interpretation of results.

Rather than presenting commands only, this book explains what each command does, why it is used, how to read the results, and how the output can be interpreted in an applied research context.

Each chapter may include a short explanation of the method, the Stata code, the Stata output, and a discussion of what the results mean.

Contents

Preface	ii
Chapter 1 Principal Component Analysis in Stata 19	1
1.1 Brief Explanation of Principal Component Analysis	1
1.2 Dataset and Variables	1
1.3 Complete Stata Do-file	2
1.4 Step-by-Step Code, Output, and Interpretation	2
1.4.1 Step 1: Load the Dataset	3
1.4.2 Step 2: Inspect the Dataset	3
1.4.3 Step 3 and Step 4: Check Missing Values and Keep Complete Observations	4
1.4.4 Step 5: Check the Correlation Matrix	4
1.4.5 Step 6: Run PCA with All Components	4
1.4.6 Step 7: Scree Plot	5
1.4.7 Step 8: Run PCA Keeping the First Two Components	6
1.4.8 Step 9: Display Component Loadings	7
1.4.9 Step 10: Generate Component Scores	7
1.4.10 Step 11: Check That the Components Are Uncorrelated	8
1.4.11 Step 12: Scatter Plot of the First Two Principal Components	8
1.4.12 Step 13: Use PC1 and PC2 as Index Variables	9
1.5 Final Summary of the PCA Result	9
1.6 Example Report Paragraph	10
Closing Note	11

Principal Component Analysis in Stata 19

Using the 1978 Automobile Dataset

1.1 Brief Explanation of Principal Component Analysis

Principal Component Analysis (PCA) is a statistical method used to reduce many related variables into a smaller number of new variables called **principal components**. These components are weighted combinations of the original variables.

The main idea is simple: if several variables move together, PCA summarizes their common pattern using fewer dimensions. For example, in a car dataset, variables such as **weight**, **length**, and **displacement** are likely to be related. Instead of analyzing all of them separately, PCA can create one component that represents the general “size and power” of a car.

PCA is useful because it helps us:

- reduce the number of variables;
- identify major patterns in the data;
- create index-like variables such as **pc1** and **pc2**;
- visualize observations using fewer dimensions.

Important terms in PCA are:

Term	Meaning
Eigenvalue	Shows how much variation a component explains. Larger eigenvalues mean more important components.
Proportion	Shows the percentage of total variation explained by each component.
Cumulative	Shows the total percentage of variation explained by components kept together.
Loadings	Show how strongly each original variable contributes to each component. Loadings help us name and interpret the components.
Scores	New variables generated for each observation, such as pc1 and pc2 . Each observation receives a score on each component.
Scree plot	A graph of eigenvalues used to decide how many components to keep.

1.2 Dataset and Variables

This example uses Stata’s built-in **1978 Automobile Data**. The PCA uses the following seven variables:

Variable	Meaning
price	Price of the car
mpg	Miles per gallon, a measure of fuel efficiency
rep78	Repair record in 1978
headroom	Interior headroom space
weight	Weight of the car
length	Length of the car
displacement	Engine displacement

Because these variables are measured in different units, Stata performs PCA using the **correlation matrix**. This standardizes the variables, so large-unit variables such as **price** or **weight** do not dominate the analysis simply because of their scale.

1.3 Complete Stata Do-file

The following do-file loads the dataset, checks the variables, runs PCA, displays loadings, generates component scores, checks that the components are uncorrelated, creates a PCA score plot, and lists the first ten observations with their PCA scores.

Stata code

```
*****
* Principal Component Analysis in Stata 19
* Dataset: 1978 Automobile Data
*****

clear all
set more off
version 19

* 1. Load dataset
webuse auto, clear

* 2. Inspect the dataset
describe
summarize price mpg rep78 headroom weight length displacement

* 3. Check missing values
misstable summarize price mpg rep78 headroom weight length displacement

* 4. Keep only complete observations for selected variables
drop if missing(price, mpg, rep78, headroom, weight, length, displacement)

* 5. Check correlation matrix
correlate price mpg rep78 headroom weight length displacement

* 6. Run PCA
pca price mpg rep78 headroom weight length displacement

* 7. Scree plot: helps decide how many components to keep
screeplot

* 8. Run PCA keeping first two components
pca price mpg rep78 headroom weight length displacement, components(2)

* 9. Display component loadings
estat loadings

* 10. Generate component scores
predict pc1 pc2, score

* 11. Check that components are uncorrelated
correlate pc1 pc2

* 12. Scatter plot of the first two principal components
scatter pc2 pc1, ///
    title("PCA Score Plot") ///
    xtitle("Principal Component 1") ///
    ytitle("Principal Component 2")

* 13. Optional: use PC1 as an index
summarize pc1 pc2
list make price mpg weight length displacement pc1 pc2 in 1/10
```

1.4 Step-by-Step Code, Output, and Interpretation

1.4.1 Step 1: Load the Dataset

Stata code

```
webuse auto, clear
```

Stata output

```
. webuse auto, clear
(1978 automobile data)
```

Interpretation

This command loads Stata's built-in automobile dataset. The output confirms that the 1978 automobile data has been successfully loaded.

1.4.2 Step 2: Inspect the Dataset

Stata code

```
describe
```

Stata output

```
. describe

Contains data from https://www.stata-press.com/data/r19/auto.dta
Observations:      74      1978 automobile data
Variables:         12      13 Apr 2024 17:45
                        (_dta has notes)
```

Variable name	Storage type	Display format	Value label	Variable label
make	str18	%-18s		Make and model
price	int	%8.0gc		Price
mpg	int	%8.0g		Mileage (mpg)
rep78	int	%8.0g		Repair record 1978
headroom	float	%6.1f		Headroom (in.)
trunk	int	%8.0g		Trunk space (cu. ft.)
weight	int	%8.0gc		Weight (lbs.)
length	int	%8.0g		Length (in.)
turn	int	%8.0g		Turn circle (ft.)
displacement	int	%8.0g		Displacement (cu. in.)
gear_ratio	float	%6.2f		Gear ratio
foreign	byte	%8.0g	origin	Car origin

Sorted by: foreign

Interpretation

The dataset has 74 observations and 12 variables. The PCA analysis will use seven numeric variables: **price**, **mpg**, **rep78**, **headroom**, **weight**, **length**, and **displacement**. The **describe** command is useful because it confirms the variable names, storage types, and variable labels before the analysis begins.

Stata code

```
summarize price mpg rep78 headroom weight length displacement
```

Stata output

```
. summarize price mpg rep78 headroom weight length displacement
```

Variable	Obs	Mean	Std. dev.	Min	Max
price	74	6165.257	2949.496	3291	15906
mpg	74	21.2973	5.785503	12	41
rep78	69	3.405797	.9899323	1	5
headroom	74	2.993243	.8459948	1.5	5
weight	74	3019.459	777.1936	1760	4840
length	74	187.9324	22.26634	142	233
displacement	74	197.2973	91.83722	79	425

Interpretation

The `summarize` command shows basic descriptive statistics. Most variables have 74 observations, but `rep78` has only 69 observations. This tells us that `rep78` has missing values. PCA should be run on a consistent sample, so we need to handle the missing observations before running PCA.

1.4.3 Step 3 and Step 4: Check Missing Values and Keep Complete Observations**Stata code**

```
misstable summarize price mpg rep78 headroom weight length displacement
drop if missing(price, mpg, rep78, headroom, weight, length, displacement)
```

Stata output

```
. misstable summarize price mpg rep78 headroom weight length displacement
```

Variable	Obs=.	Obs>.	Obs<.	Unique values	Min	Max
rep78	5		69	5	1	5

```
. drop if missing(price, mpg, rep78, headroom, weight, length, displacement)
(5 observations deleted)
```

Interpretation

The missing-value table shows that `rep78` has 5 missing observations. The `drop if missing(...)` command removes those incomplete observations. After this step, 69 observations remain. This ensures that every car used in PCA has complete information for all seven variables.

1.4.4 Step 5: Check the Correlation Matrix**Stata code**

```
correlate price mpg rep78 headroom weight length displacement
```

Stata output

```
. correlate price mpg rep78 headroom weight length displacement
(obs=69)
```

	price	mpg	rep78	headroom	weight	length	displacement
price	1.0000						
mpg	-0.4559	1.0000					
rep78	0.0066	0.4023	1.0000				
headroom	0.1112	-0.3996	-0.1480	1.0000			
weight	0.5478	-0.8055	-0.4003	0.4795	1.0000		
length	0.4425	-0.8037	-0.3606	0.5240	0.9478	1.0000	
displacement	0.5479	-0.7434	-0.4119	0.4763	0.9316	0.8621	1.0000

Interpretation

The correlation matrix shows that PCA is appropriate because several variables are strongly related. For example, `weight` and `length` are highly positively correlated, and `weight` and `displacement` are also highly positively correlated. Meanwhile, `mpg` is strongly negatively correlated with `weight`, `length`, and `displacement`. This makes sense: larger and heavier cars usually have lower fuel efficiency.

1.4.5 Step 6: Run PCA with All Components**Stata code**

```
pca price mpg rep78 headroom weight length displacement
```

Stata output

```
Principal components/correlation      Number of obs      =      69
                                     Number of comp.    =      7
                                     Trace                =      7
                                     Rho                  =      1.0000

      Rotation: (unrotated = principal)
```

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	4.34052	3.32161	0.6201	0.6201
Comp2	1.01891	.184366	0.1456	0.7656
Comp3	.834546	.443705	0.1192	0.8849
Comp4	.390842	.116964	0.0558	0.9407
Comp5	.273877	.162216	0.0391	0.9798
Comp6	.111662	.0820227	0.0160	0.9958
Comp7	.0296392	.	0.0042	1.0000

Principal components (eigenvectors)

Variable	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6	Comp7	Unexplained
price	0.2761	0.6781	-0.2652	0.5810	-0.1570	0.1653	0.0632	0
mpg	-0.4186	0.0202	0.1017	0.3700	0.7906	0.2281	0.0050	0
rep78	-0.2222	0.7039	0.4923	-0.4419	0.0433	-0.1222	-0.0259	0
headroom	0.2713	-0.2016	0.8172	0.4367	-0.1624	0.0068	-0.0293	0
weight	0.4660	0.0442	-0.0304	-0.1611	0.2893	0.1408	-0.8065	0
length	0.4525	-0.0128	0.0808	-0.3368	0.2070	0.6132	0.5063	0
displacement	0.4513	0.0388	-0.0420	0.0116	0.4421	-0.7141	0.2961	0

Interpretation

The PCA creates seven components because seven variables were used. When all seven components are retained, $Rho = 1.0000$, meaning all components together explain 100% of the total variation. The first component has an eigenvalue of 4.34052 and explains 62.01% of the total variation. The second component has an eigenvalue of 1.01891 and explains another 14.56%. Together, the first two components explain 76.56% of the variation. The first component is the most important because it explains the largest share of total variation.

1.4.6 Step 7: Scree Plot**Stata code**

```
screeplot
```

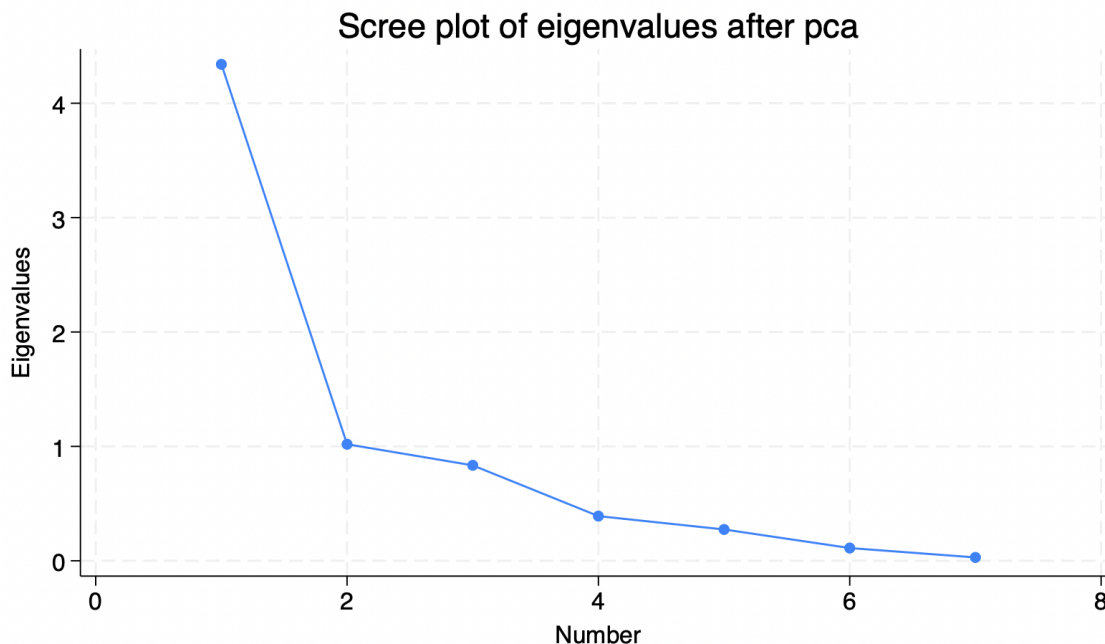


Figure 1.1: Scree plot of eigenvalues after PCA

Interpretation

The scree plot shows eigenvalues on the vertical axis and component numbers on the horizontal axis. The plot drops sharply from Component 1 to Component 2 and then becomes much flatter. This suggests that the first two components contain most of the useful information. The eigenvalue rule also supports keeping two components because Component 1 and Component 2 have eigenvalues around or above 1, while Component 3 and later components are below 1.

1.4.7 Step 8: Run PCA Keeping the First Two Components

Stata code

```
pca price mpg rep78 headroom weight length displacement, components(2)
```

Stata output

```
Principal components/correlation      Number of obs   =      69
                                     Number of comp. =       2
                                     Trace              =       7
                                     Rho              =    0.7656

Rotation: (unrotated = principal)
```

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	4.34052	3.32161	0.6201	0.6201
Comp2	1.01891	.184366	0.1456	0.7656
Comp3	.834546	.443705	0.1192	0.8849
Comp4	.390842	.116964	0.0558	0.9407
Comp5	.273877	.162216	0.0391	0.9798
Comp6	.111662	.0820227	0.0160	0.9958
Comp7	.0296392	.	0.0042	1.0000

Principal components (eigenvectors)

Variable	Comp1	Comp2	Unexplained
price	0.2761	0.6781	.2006
mpg	-0.4186	0.0202	.2391
rep78	-0.2222	0.7039	.2808
headroom	0.2713	-0.2016	.6391
weight	0.4660	0.0442	.05533
length	0.4525	-0.0128	.1111
displacement	0.4513	0.0388	.1146

Interpretation

This command keeps only the first two principal components. Therefore, $Rho = 0.7656$, meaning the first two components together explain 76.56% of the total variation. This is a strong reduction because seven original variables are summarized into two components.

The **Unexplained** column shows how much of each variable is not explained by the two retained components. For example, only 5.533% of **weight** remains unexplained, so PC1 and PC2 explain weight very well. However, 63.91% of **headroom** remains unexplained, meaning headroom is not captured as well by the first two components.

1.4.8 Step 9: Display Component Loadings**Stata code**

```
estat loadings
```

Stata output

Principal component loadings (unrotated)
component normalization: sum of squares(column) = 1

	Comp1	Comp2
price	.2761	.6781
mpg	-.4186	.02018
rep78	-.2222	.7039
headroom	.2713	-.2016
weight	.4660	.04421
length	.4525	-.01278
displacement	.4513	.03882

Interpretation

Component loadings show the contribution of each original variable to each component. Large positive or negative values are the most important.

For **Component 1**, the strongest positive loadings are **weight** (.4660), **length** (.4525), and **displacement** (.4513). The strongest negative loading is **mpg** (-.4186). Therefore, Component 1 can be interpreted as a **car size/power component**: higher PC1 scores indicate larger, heavier, higher-displacement cars with lower fuel efficiency.

For **Component 2**, the strongest positive loadings are **price** (.6781) and **rep78** (.7039). Therefore, Component 2 can be interpreted as a **price/reliability component**: higher PC2 scores indicate more expensive cars with better repair records.

1.4.9 Step 10: Generate Component Scores**Stata code**

```
predict pc1 pc2, score
```

Stata output

Scoring coefficients
sum of squares(column-loading) = 1

	Comp1	Comp2
price	0.2761	0.6781
mpg	-0.4186	0.0202
rep78	-0.2222	0.7039
headroom	0.2713	-0.2016
weight	0.4660	0.0442
length	0.4525	-0.0128
displacement	0.4513	0.0388

Interpretation

The command `predict pc1 pc2, score` creates two new variables in the dataset: `pc1` and `pc2`. Each car now receives a score on Component 1 and Component 2. These scores can be used as index variables.

A high `pc1` score means the car is likely to be large, heavy, powerful, and less fuel-efficient. A high `pc2` score means the car is likely to be more expensive and have a better repair record.

1.4.10 Step 11: Check That the Components Are Uncorrelated**Stata code**

```
correlate pc1 pc2
```

Stata output

```
. correlate pc1 pc2
(obs=69)
```

	pc1	pc2
pc1	1.0000	
pc2	0.0000	1.0000

Interpretation

The correlation between `pc1` and `pc2` is 0.0000. This confirms one of the main properties of PCA: principal components are uncorrelated with each other. Component 1 captures the largest pattern in the data, while Component 2 captures a different pattern that does not overlap with Component 1.

1.4.11 Step 12: Scatter Plot of the First Two Principal Components**Stata code**

```
scatter pc2 pc1, ///
    title("PCA Score Plot") ///
    xtitle("Principal Component 1") ///
    ytitle("Principal Component 2")
```

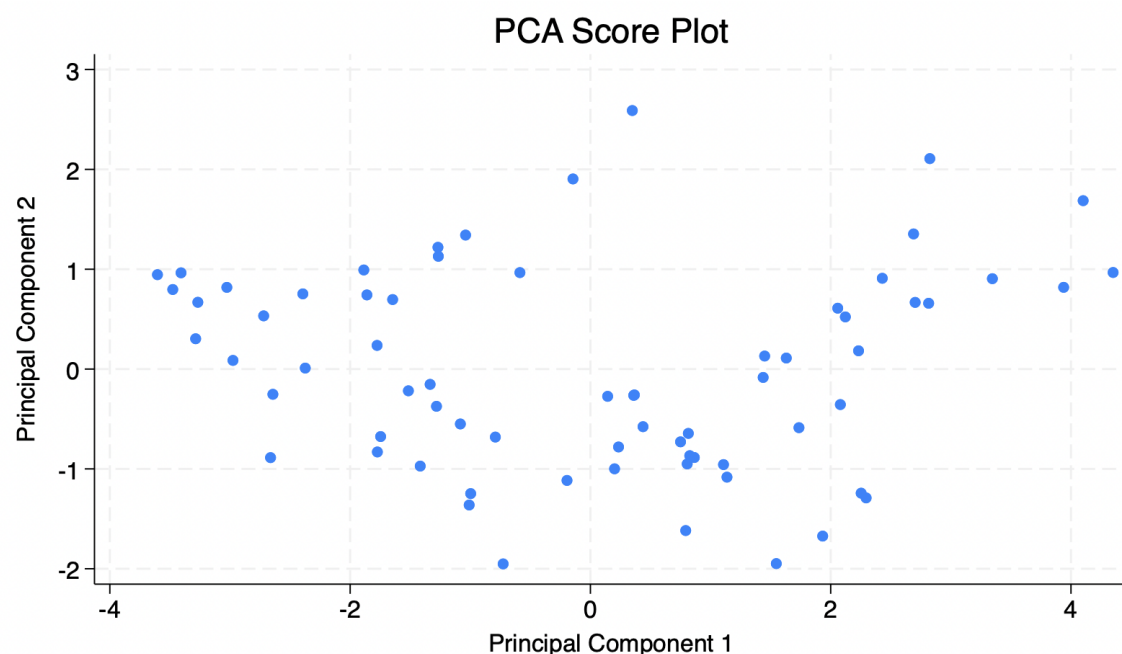


Figure 1.2: PCA score plot using the first two principal components

Interpretation

The PCA score plot places each car on a two-dimensional graph. The horizontal axis is PC1 and the vertical axis is PC2. Each point represents one car.

Cars on the right side of the plot have higher PC1 scores, meaning they tend to be larger, heavier, more powerful, and less fuel-efficient. Cars on the left side tend to be smaller and more fuel-efficient. Cars higher on the graph have higher PC2 scores, meaning they tend to be more expensive and have better repair records. Cars lower on the graph tend to be cheaper or have weaker repair records.

Therefore, the score plot visually summarizes the automobile dataset using two dimensions: **size/power** on the horizontal axis and **price/reliability** on the vertical axis.

1.4.12 Step 13: Use PC1 and PC2 as Index Variables

Stata code

```
summarize pc1 pc2

list make price mpg weight length displacement pc1 pc2 in 1/10
```

Stata output

```
. summarize pc1 pc2
```

Variable	Obs	Mean	Std. dev.	Min	Max
pc1	69	1.27e-08	2.083392	-3.603725	4.350894
pc2	69	-4.10e-09	1.009412	-1.951135	2.589266

```
. list make price mpg weight length displacement pc1 pc2 in 1/10
```

	make	price	mpg	weight	length	displacement	pc1	pc2
1.	AMC Concord	4,099	22	2,930	186	121	-.7911875	-.6810844
2.	AMC Pacer	4,749	17	3,350	173	258	.4381556	-.5772535
3.	Buick Century	4,816	20	3,250	196	196	.8058355	-.9500939
4.	Buick Electra	7,827	15	4,080	222	350	2.815666	.6588394
5.	Buick LeSabre	5,788	18	3,670	218	231	1.735743	-.5868738
6.	Buick Regal	5,189	20	3,280	200	196	.1434062	-.2731262
7.	Buick Riviera	10,372	16	3,880	207	231	2.05864	.6095829
8.	Buick Skylark	4,082	19	3,400	200	231	.8268985	-.8674498
9.	Cad. Deville	11,385	14	4,330	221	425	3.939213	.8185096
10.	Cad. Eldorado	14,500	14	3,900	204	350	3.345759	.9051669

Interpretation

The summary statistics show that **pc1** and **pc2** have means very close to zero. That is normal because PCA scores are centered around zero. The standard deviation of **pc1** is larger than the standard deviation of **pc2**, which is consistent with PC1 explaining more variation than PC2.

The **list** output shows the first ten cars with their original variables and component scores. For example, **Cad. Deville** has a high **pc1** score of 3.939213. This is consistent with its high weight, long length, large displacement, and low miles per gallon. **Cad. Eldorado** also has a high **pc1** score and a positive **pc2** score, suggesting that it is large/powerful and relatively expensive/reliable according to the PCA dimensions.

1.5 Final Summary of the PCA Result

The PCA shows that the seven automobile variables can be summarized well using the first two principal components. Component 1 explains 62.01% of the total variation and mainly represents car size and power, because it loads strongly on **weight**, **length**, and **displacement**, while loading negatively on **mpg**. Component 2 explains another 14.56% of the variation and mainly represents price and reliability, because it loads strongly on **price** and **rep78**. Together, the first two components explain 76.56% of the total variation.

A concise interpretation is:

Component	Interpretation
PC1	Car size/power versus fuel efficiency
PC2	Price/reliability or price-quality dimension

Therefore, instead of analyzing seven original variables separately, we can use `pc1` and `pc2` as two summary indexes for the main patterns in the automobile data.

1.6 Example Report Paragraph

Ready-to-use paragraph

Principal Component Analysis was conducted using seven automobile characteristics: price, miles per gallon, repair record, headroom, weight, length, and displacement. The first principal component had an eigenvalue of 4.34052 and explained 62.01% of the total variation. The second component had an eigenvalue of 1.01891 and explained 14.56% of the variation. Together, the first two components explained 76.56% of the total variation. The loading matrix shows that the first component loads strongly on weight, length, and displacement, while loading negatively on miles per gallon; therefore, it can be interpreted as a car size/power component. The second component loads strongly on price and repair record, suggesting a price/reliability component. The generated component scores, `pc1` and `pc2`, are uncorrelated, confirming that they capture separate dimensions of variation in the automobile data.

Closing Note

This book is designed to grow over time. Every new Stata topic can become a new chapter or appendix.